

# Supplementary Material for “Exact Structural Abstraction and Tractability Limits”

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## 1 Full Lean Handle Ledger

This supplement provides the complete Lean handle ledger cited by the manuscript. It includes every handle identifier, declaration name, and source module path.

| ID   | Lean Handle / Source                                                              | ID   | Lean Handle / Source                                                                            |
|------|-----------------------------------------------------------------------------------|------|-------------------------------------------------------------------------------------------------|
| FR1  | currentFamilies_partition_by_role<br>Paper4dFrontier/Classification.lean          | FR2  | currentFamilies_partition_counts<br>Paper4dFrontier/Classification.lean                         |
| FR3  | current_families_have_finite_basis<br>Paper4dFrontier/Classification.lean         | FR4  | current_positive_interfaces_factor_through_role<br>Paper4dFrontier/Classification.lean          |
| FR5  | lifted_families_reduce_to_core<br>Paper4dFrontier/Classification.lean             | FR6  | degenerateFamilies_complete<br>Paper4dFrontier/Classification.lean                              |
| FR7  | exists_decisionProblem_realizing_labeling<br>Paper4dFrontier/Realizability.lean   | FR8  | realizingProblem_decisionEquiv_iff<br>Paper4dFrontier/Realizability.lean                        |
| FR13 | isOptimal_positiveAffineTransform_iff<br>Paper4dFrontier/FamilyAxioms.lean        | FR14 | opt_eq_positiveAffineTransform<br>Paper4dFrontier/FamilyAxioms.lean                             |
| FR15 | isSufficient_positiveAffineTransform_iff<br>Paper4dFrontier/FamilyAxioms.lean     | FR16 | isRelevant_positiveAffineTransform_iff<br>Paper4dFrontier/FamilyAxioms.lean                     |
| FR17 | decisionEquiv_relabelActions_iff<br>Paper4dFrontier/FamilyAxioms.lean             | FR18 | decisionEquiv_relabelStates_iff<br>Paper4dFrontier/FamilyAxioms.lean                            |
| FR19 | isSufficient_relabelActions_iff<br>Paper4dFrontier/FamilyAxioms.lean              | FR20 | isSufficient_relabelStates_iff<br>Paper4dFrontier/FamilyAxioms.lean                             |
| FR21 | allProblems_relabelInvariant<br>Paper4dFrontier/FamilyAxioms.lean                 | FR22 | allProblems_kernelUniversal<br>Paper4dFrontier/FamilyAxioms.lean                                |
| FR23 | relabelInvariant_not_enough<br>Paper4dFrontier/FamilyAxioms.lean                  | FR24 | optRangeCard_realizingIdentity<br>Paper4dFrontier/FamilyAxioms.lean                             |
| FR25 | currentFamilies_explicit<br>Paper4dFrontier/Classification.lean                   | FR26 | coreFamilies_explicit<br>Paper4dFrontier/Classification.lean                                    |
| FR27 | liftedFamilies_explicit<br>Paper4dFrontier/Classification.lean                    | FR28 | degenerateFamilies_explicit<br>Paper4dFrontier/Classification.lean                              |
| FR29 | primitiveMechanisms_explicit<br>Paper4dFrontier/Classification.lean               | FR30 | isSufficient_iff_of_decisionEquiv_iff<br>Paper4dFrontier/FamilyAxioms.lean                      |
| FR31 | isRelevant_iff_of_decisionEquiv_iff<br>Paper4dFrontier/FamilyAxioms.lean          | FR32 | isIrrelevant_iff_of_decisionEquiv_iff<br>Paper4dFrontier/FamilyAxioms.lean                      |
| FR33 | isMinimalSufficient_iff_of_decisionEquiv_iff<br>Paper4dFrontier/FamilyAxioms.lean | FR34 | sufficientSets_eq_of_decisionEquiv_iff<br>Paper4dFrontier/FamilyAxioms.lean                     |
| FR35 | relevantSet_eq_of_decisionEquiv_iff<br>Paper4dFrontier/FamilyAxioms.lean          | FR37 | quotientCard_eq_of_decisionEquiv_iff<br>Paper4dFrontier/FamilyAxioms.lean                       |
| FR38 | quotientCard_realizingIdentity<br>Paper4dFrontier/FamilyAxioms.lean               | FR39 | realizingProblemQuotientEquivLabelRange_apply_quotientMap<br>Paper4dFrontier/Realizability.lean |

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| ID    | Lean Handle / Source                                                                                              | ID    | Lean Handle / Source                                                                                             |
|-------|-------------------------------------------------------------------------------------------------------------------|-------|------------------------------------------------------------------------------------------------------------------|
| FR40  | realizingProblem_quotientCard_eq_labelRangeCard<br>Paper4dFrontier/Realizability.lean                             | FR41  | certificationStatistic_eq_of_decisionEquiv_iff<br>Paper4dFrontier/FamilyAxioms.lean                              |
| FR42  | sufficientSetCount_eq_of_decisionEquiv_iff<br>Paper4dFrontier/FamilyAxioms.lean                                   | FR43  | minimalSufficientSetCount_eq_of_decisionEquiv_iff<br>Paper4dFrontier/FamilyAxioms.lean                           |
| FR44  | relevantCoordCount_eq_of_decisionEquiv_iff<br>Paper4dFrontier/FamilyAxioms.lean                                   | FR45  | setoidRealizingProblem_decisionEquiv_iff<br>Paper4dFrontier/Realizability.lean                                   |
| FR46  | setoidRealizingProblemQuotientEquivSetoidQuotient_apply_quotientMap<br>Paper4dFrontier/Realizability.lean         | FR47  | exists_decisionProblem_realizing_setoid<br>Paper4dFrontier/Realizability.lean                                    |
| FR48  | isSufficient_iff_agreementRel_le_decisionEquiv<br>Paper4dFrontier/FamilyAxioms.lean                               | FR49  | isIrrelevant_iff_sufficient_erase<br>Paper4dFrontier/FamilyAxioms.lean                                           |
| FR50  | isRelevant_iff_not_sufficient_erase<br>Paper4dFrontier/FamilyAxioms.lean                                          | FR51  | quotientEquiv_of_decisionEquiv_iff_apply_quotientMap<br>Paper4dFrontier/FamilyAxioms.lean                        |
| FR52  | empty_sufficient_of_constant_opt<br>Paper4dFrontier/ClosureLaws.lean                                              | FR53  | irrelevant_of_constant_opt<br>Paper4dFrontier/ClosureLaws.lean                                                   |
| FR54  | decisionEquiv_duplicateAction_iff<br>Paper4dFrontier/ClosureLaws.lean                                             | FR55  | isSufficient_duplicateAction_iff<br>Paper4dFrontier/ClosureLaws.lean                                             |
| FR56  | isRelevant_duplicateAction_iff<br>Paper4dFrontier/ClosureLaws.lean                                                | FR57  | decisionEquiv_duplicateState_iff<br>Paper4dFrontier/ClosureLaws.lean                                             |
| FR58  | isSufficient_duplicateState_iff<br>Paper4dFrontier/ClosureLaws.lean                                               | FR59  | isRelevant_duplicateState_iff<br>Paper4dFrontier/ClosureLaws.lean                                                |
| FR60  | duplicateStateQuotientEquivOriginal_apply_quotientMap<br>Paper4dFrontier/ClosureLaws.lean                         | FR66  | booleanCube_card<br>Paper4dFrontier/EnumerationBounds.lean                                                       |
| FR67  | exists_booleanCube_larger_than_monomial<br>Paper4dFrontier/EnumerationBounds.lean                                 | FR68  | decisionEquiv_addDuplicateState_iff<br>Paper4dFrontier/ClosureLaws.lean                                          |
| FR69  | isSufficient_addDuplicateState_iff<br>Paper4dFrontier/ClosureLaws.lean                                            | FR70  | isRelevant_addDuplicateState_iff<br>Paper4dFrontier/ClosureLaws.lean                                             |
| FR71  | addDuplicateStateQuotientEquivOriginal_apply_quotientMap<br>Paper4dFrontier/ClosureLaws.lean                      | FR72  | some_mem_opt_addDuplicateAction_iff<br>Paper4dFrontier/ClosureLaws.lean                                          |
| FR73  | none_mem_opt_addDuplicateAction_iff<br>Paper4dFrontier/ClosureLaws.lean                                           | FR83  | isSufficient_noiseExtended_iff<br>Paper4dFrontier/DimensionalNoiseExtension.lean                                 |
| FR84  | isRelevant_noiseExtended_iff<br>Paper4dFrontier/DimensionalNoiseExtension.lean                                    | FR85  | lastCoord_irrelevant_noiseExtended<br>Paper4dFrontier/DimensionalNoiseExtension.lean                             |
| FR100 | parentTree_realTreewidth_one<br>Paper4dFrontier/ParentTreewidth.lean                                              | FR101 | low_rank_only_witness<br>Paper4dFrontier/LowRankOnlyWitness.lean                                                 |
| FR105 | separable_only_witness<br>Paper4dFrontier/SeparableOnlyWitness.lean                                               | FR106 | bounded_actions_only_witness<br>Paper4dFrontier/BoundedActionsOnlyWitness2.lean                                  |
| FR108 | all_independent_core_mechanisms_nondegenerate<br>Paper4dFrontier/Block2Progress.lean                              | FR110 | parentTreeStructured_implies_treeStructured<br>Paper4dFrontier/ParentTreewidth.lean                              |
| FR111 | treeStructured_and_unique_parent_iff_parentTreeStructured<br>Paper4dFrontier/ParentTreewidth.lean                 | FR112 | bounded_treewidth_only_witness<br>Paper4dFrontier/BoundedTreewidthOnlyWitness.lean                               |
| FR113 | symmetry_only_witness<br>Paper4dFrontier/SymmetryOnlyWitness.lean                                                 | FR114 | weak_tree_structured_not_width_one<br>Paper4dFrontier/LegacyTreeGap.lean                                         |
| FR115 | tree_structure_hardness_bridge<br>Paper4dFrontier/TreeStructureHardnessBridge.lean                                | FR118 | pairCrossDifference_eq_binaryCrossDifference_of_lt<br>Paper4dFrontier/BinaryPairwiseDichotomy.lean               |
| FR119 | pairwise_zero_crossDifference_unaryDecomposition<br>Paper4dFrontier/BinaryPairwiseDichotomy.lean                  | FR120 | binary_pairwise_symmetry_dichotomy<br>Paper4dFrontier/BinaryPairwiseDichotomy.lean                               |
| FR121 | actionGapCrossDifference_eq_binaryCrossDifference_of_lt<br>Paper4dFrontier/DecisionRelevantPairwiseDichotomy.lean | FR122 | decisionRelevant_zero_actionGap_implies_unaryReduction<br>Paper4dFrontier/DecisionRelevantPairwiseDichotomy.lean |
| FR123 | binary_pairwise_symmetry_decision_relevant_dichotomy<br>Paper4dFrontier/DecisionRelevantPairwiseDichotomy.lean    | FR124 | block6_genuine_interaction_dichotomy_obstruction<br>Paper4dFrontier/Block6Obstruction.lean                       |

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| ID    | Lean Handle / Source                                                                                                                | ID    | Lean Handle / Source                                                                                                                 |
|-------|-------------------------------------------------------------------------------------------------------------------------------------|-------|--------------------------------------------------------------------------------------------------------------------------------------|
| FR125 | decisionRelevantInteractionGraph_<br>addActionOffset<br>Paper4dFrontier/DecisionRelevantPairwiseDichotomy.lean                      | FR126 | action_offset_can_force_constant_optimizer<br>Paper4dFrontier/Block6Obstruction.lean                                                 |
| FR127 | block6_action_offset_obstruction<br>Paper4dFrontier/Block6Obstruction.lean                                                          | FR128 | offsetNormalizedDecisionRelevantInteractionGraph_<br>wellDefined<br>Paper4dFrontier/DecisionRelevantPairwiseDichotomy.lean           |
| FR129 | block6_offset_normalized_obstruction<br>Paper4dFrontier/Block6Obstruction.lean                                                      | FR130 | neverOptimalGhost_decisionRelevantGraph_eq_top<br>Paper4dFrontier/Block6Obstruction.lean                                             |
| FR131 | neverOptimalGhost_supportedGraph_eq_bot<br>Paper4dFrontier/Block6Obstruction.lean                                                   | FR132 | support_filtering_removes_known_block6_<br>obstructions<br>Paper4dFrontier/Block6Obstruction.lean                                    |
| FR133 | block6_ghost_action_obstruction<br>Paper4dFrontier/Block6Obstruction.lean                                                           | FR134 | marginMasking_supportedGraph_eq_top<br>Paper4dFrontier/Block6Obstruction.lean                                                        |
| FR135 | marginMasking_zero_sufficient<br>Paper4dFrontier/Block6Obstruction.lean                                                             | FR136 | block6_optimizer_supported_obstruction<br>Paper4dFrontier/Block6Obstruction.lean                                                     |
| FR137 | MarginBounded<br>Paper4dFrontier/DecisionRelevantPairwiseDichotomy.lean                                                             | FR138 | dominantPair_marginBounded<br>Paper4dFrontier/Block6Obstruction.lean                                                                 |
| FR139 | dominantPair_supportedGraph_eq_top<br>Paper4dFrontier/Block6Obstruction.lean                                                        | FR140 | block6_margin_bounded_obstruction<br>Paper4dFrontier/Block6Obstruction.lean                                                          |
| FR145 | not_collapseLandscapeInfinity_of_oracle_<br>predicate<br>Paper4dFrontier/MetaCharacterization.lean                                  | FR176 | DecisionQuotient.DecisionProblem.quotient_is_<br>coarsest<br>paper4/DecisionQuotient/Quotient.lean                                   |
| FR177 | DecisionQuotient.DecisionProblem.quotient_has_<br>unique_factorization<br>paper4/DecisionQuotient/Quotient.lean                     | FR178 | DecisionQuotient.DecisionProblem.srank_eq_<br>relevant_card<br>paper4/DecisionQuotient/Tractability/<br>StructuralRank.lean          |
| FR179 | DecisionQuotient.quotientEntropy_le_srank_<br>binary<br>paper4/DecisionQuotient/Information.lean                                    | FR180 | DecisionQuotient.Statistics.fisherMatrix_rank_<br>eq_srank<br>paper4/DecisionQuotient/Statistics/<br>FisherInformation.lean          |
| FR181 | DecisionQuotient.Information.compression_<br>below_srank_fails<br>paper4/DecisionQuotient/Information/<br>RDSrank.lean              | FR182 | DecisionQuotient.Information.srank_bits_<br>sufficient<br>paper4/DecisionQuotient/Information/<br>RDSrank.lean                       |
| FR183 | DecisionQuotient.Physics.single_future_zero_<br>cost<br>paper4/DecisionQuotient/Physics/<br>WassersteinIntegrity.lean               | FR184 | DecisionQuotient.Physics.transportCost_pos_of_<br>offDiag<br>paper4/DecisionQuotient/Physics/<br>WassersteinIntegrity.lean           |
| FR185 | no_closureInvariant_predicate_of_orbit_gap<br>Paper4dFrontier/ObstructionPredicateCandidates.lean                                   | FR186 | no_admissibleNormalizationPredicate_decides_<br>dominantPair<br>Paper4dFrontier/ObstructionPredicateCandidates.lean                  |
| FR187 | no_admissibleNormalizationPredicate_decides_<br>marginBounded<br>Paper4dFrontier/ObstructionPredicateCandidates.lean                | FR188 | no_admissibleNormalizationPredicate_decides_<br>ghostActionTwoPairCrossOne<br>Paper4dFrontier/ObstructionPredicateCandidates.lean    |
| FR189 | no_admissibleNormalizationPredicate_decides_<br>offsetActionZeroPairCrossOne<br>Paper4dFrontier/ObstructionPredicateCandidates.lean | FR190 | admissibleCollapseLandscapeInfinity_full_paper<br>Paper4dFrontier/ObstructionPredicateCandidates.lean                                |
| FR191 | ClosureSoundPackage<br>Paper4dFrontier/ObstructionPredicateCandidates.lean                                                          | FR192 | ReasonableGuardrailPackage<br>Paper4dFrontier/ObstructionPredicateCandidates.lean                                                    |
| FR193 | no_closureSoundPackage_predicate_decides_four_<br>obstruction_families<br>Paper4dFrontier/ObstructionPredicateCandidates.lean       | FR194 | no_reasonableGuardrailPackage_predicate_<br>decides_four_obstruction_families<br>Paper4dFrontier/ObstructionPredicateCandidates.lean |
| FR197 | classifier_agrees_on_closureEquivalent_of_<br>correctOnDomain<br>Paper4dFrontier/AdmissibleCharacterization.lean                    | FR198 | no_correctOnDomain_classifier_of_orbit_gap<br>Paper4dFrontier/AdmissibleCharacterization.lean                                        |
| FR199 | correct_classifier_inherits_<br>closureLawInvariant<br>Paper4dFrontier/AdmissibleCharacterization.lean                              | FR200 | admissibleNormalizationPredicate_has_explicit_<br>inhabitants<br>Paper4dFrontier/AdmissibleCharacterization.lean                     |
| FR201 | closureLawInvariant_of_iff_of_<br>closureEquivalent<br>Paper4dFrontier/AdmissibleCharacterization.lean                              | FR202 | exists_orbit_gap_of_not_closureLawInvariant<br>Paper4dFrontier/AdmissibleCharacterization.lean                                       |

(continued...)

| ID    | Lean Handle / Source                                                                                                                             | ID    | Lean Handle / Source                                                                                                                              |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| FR203 | <code>closureLawInvariant_iff_no_orbit_gap</code><br><code>Paper4dFrontier/AdmissibleCharacterization.lean</code>                                | FR204 | <code>no_exact_closureLawInvariant_classifier_iff_exists_orbit_gap</code><br><code>Paper4dFrontier/AdmissibleCharacterization.lean</code>         |
| FR205 | <code>distinctActionCount</code><br><code>Paper4dFrontier/DistinctActionProfiles.lean</code>                                                     | FR206 | <code>actionCount_profileCompressedSlice</code><br><code>Paper4dFrontier/DistinctActionProfiles.lean</code>                                       |
| FR207 | <code>decisionEquiv_profileCompressedSlice_iff</code><br><code>Paper4dFrontier/DistinctActionProfiles.lean</code>                                | FR208 | <code>profileCompressedSlice_preserves_exactCertification</code><br><code>Paper4dFrontier/DistinctActionProfiles.lean</code>                      |
| FR209 | <code>profileCompressedSlice_bounded_actions</code><br><code>Paper4dFrontier/DistinctActionProfiles.lean</code>                                  | FR210 | <code>OrbitGapOn</code><br><code>Paper4dFrontier/AdmissibleCharacterization.lean</code>                                                           |
| FR211 | <code>no_orbitGapOn_of_exact_classifiable_by_closureLawInvariant_onDomain</code><br><code>Paper4dFrontier/AdmissibleCharacterization.lean</code> | FR212 | <code>exact_classifiable_by_closureLawInvariant_onDomain_iff_no_orbitGapOn</code><br><code>Paper4dFrontier/AdmissibleCharacterization.lean</code> |
| FR213 | <code>no_exact_closureLawInvariant_classifier_onDomain_iff_orbitGapOn</code><br><code>Paper4dFrontier/AdmissibleCharacterization.lean</code>     | FR214 | <code>boundedPatternScheme_holds_largeActionCount_iff</code><br><code>Paper4dFrontier/AdmissibleCharacterization.lean</code>                      |
| FR215 | <code>boundedPatternDefinable_eventually_constant_in_actionCount</code><br><code>Paper4dFrontier/AdmissibleCharacterization.lean</code>          | FR216 | <code>boundedPatternDefinable_largeActionCount_agrees</code><br><code>Paper4dFrontier/AdmissibleCharacterization.lean</code>                      |
| FR217 | <code>payloadSufficient_iff_realizingProblem_isSufficient</code><br><code>Paper4dFrontier/Realizability.lean</code>                              | FR218 | <code>payloadRelevant_iff_realizingProblem_isRelevant</code><br><code>Paper4dFrontier/Realizability.lean</code>                                   |
| FR219 | <code>payloadIrrelevant_iff_realizingProblem_isIrrelevant</code><br><code>Paper4dFrontier/Realizability.lean</code>                              | FR220 | <code>payloadMinimalSufficient_iff_realizingProblem_isMinimalSufficient</code><br><code>Paper4dFrontier/Realizability.lean</code>                 |
| FR221 | <code>payloadSufficientSets_eq_realizingProblem_sufficientSets</code><br><code>Paper4dFrontier/Realizability.lean</code>                         | FR222 | <code>payloadRelevantSet_eq_realizingProblem_relevantSet</code><br><code>Paper4dFrontier/Realizability.lean</code>                                |
| FR223 | <code>feasiblePayloadSufficient_iff_lawDecisionProblem_isSufficient</code><br><code>Paper4dFrontier/Realizability.lean</code>                    | FR224 | <code>feasiblePayloadRelevant_iff_lawDecisionProblem_isRelevant</code><br><code>Paper4dFrontier/Realizability.lean</code>                         |
| FR225 | <code>feasiblePayloadIrrelevant_iff_lawDecisionProblem_isIrrelevant</code><br><code>Paper4dFrontier/Realizability.lean</code>                    | FR226 | <code>setValuedPayloadSufficient_iff_totalizedLawDecisionProblem_isSufficient</code><br><code>Paper4dFrontier/Realizability.lean</code>           |
| FR227 | <code>setValuedPayloadRelevant_iff_totalizedLawDecisionProblem_isRelevant</code><br><code>Paper4dFrontier/Realizability.lean</code>              | FR228 | <code>setValuedPayloadIrrelevant_iff_totalizedLawDecisionProblem_isIrrelevant</code><br><code>Paper4dFrontier/Realizability.lean</code>           |
| FR229 | <code>outputSemanticsSufficient_iff_totalizedLawDecisionProblem_isSufficient</code><br><code>Paper4dFrontier/Realizability.lean</code>           | FR230 | <code>outputSemanticsRelevant_iff_totalizedLawDecisionProblem_isRelevant</code><br><code>Paper4dFrontier/Realizability.lean</code>                |
| FR231 | <code>outputSemanticsIrrelevant_iff_totalizedLawDecisionProblem_isIrrelevant</code><br><code>Paper4dFrontier/Realizability.lean</code>           | FR232 | <code>approximationSemanticsSufficient_iff_totalizedLawDecisionProblem_isSufficient</code><br><code>Paper4dFrontier/Realizability.lean</code>     |
| FR233 | <code>approximationSemanticsRelevant_iff_totalizedLawDecisionProblem_isRelevant</code><br><code>Paper4dFrontier/Realizability.lean</code>        | FR234 | <code>approximationSemanticsIrrelevant_iff_totalizedLawDecisionProblem_isIrrelevant</code><br><code>Paper4dFrontier/Realizability.lean</code>     |
| FR235 | <code>outputSemanticsSufficientSets_eq_of_outputSemanticsEquivalent</code><br><code>Paper4dFrontier/Realizability.lean</code>                    | FR236 | <code>outputSemanticsRelevantSet_eq_of_outputSemanticsEquivalent</code><br><code>Paper4dFrontier/Realizability.lean</code>                        |
| FR237 | <code>quotientMap_realizes_setoid_asOutputSemantics</code><br><code>Paper4dFrontier/Realizability.lean</code>                                    | FR240 | <code>exists_outputSemantics_realizing_setoid</code><br><code>Paper4dFrontier/Realizability.lean</code>                                           |

## 2 Claim-to-Lean Handle Mapping

This section maps each paper claim to its corresponding Lean formalization.

| Paper claim                                                             | Lean handle         |
|-------------------------------------------------------------------------|---------------------|
| Corollary 6.25: Exact Approximation Semantics Transfer                  | FR234, FR233, FR232 |
| Corollary 6.6: Constant-Optimizer Collapse Forces Trivial Certification | FR52, FR53          |

| Paper claim                                                                              | Lean handle                                                                              |
|------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Corollary 6.27: Every State Equivalence Relation Is Realizable as Exact Output Semantics | FR240, FR237                                                                             |
| Corollary 2.3: Finite Basis for the Current Positive Landscape                           | FR3, FR29                                                                                |
| Corollary 4.11: No Correct Tractability Classifier                                       | FR197, FR199, FR193, FR198                                                               |
| Corollary 4.13: Orbit-Gap Completeness on Closure-Closed Domains                         | FR210, FR212, FR213, FR211                                                               |
| Corollary 5.4: Realizability Alone Cannot Be the Frontier                                | FR7, FR8                                                                                 |
| Corollary 4.10: Reasonable Guardrail Packages                                            | FR194                                                                                    |
| Corollary 6.24: Arbitrary Exact Output Semantics Transfer                                | FR231, FR230, FR229                                                                      |
| Corollary 6.3: Relevance Is Failure of Erased Sufficiency                                | FR49, FR50                                                                               |
| Corollary 6.22: Nonempty Set-Valued Payload Transfer                                     | FR225, FR224, FR223                                                                      |
| Corollary 6.4: Certification Statistics Factor Through the Quotient Relation             | FR41, FR43, FR44, FR42                                                                   |
| Corollary 6.2: Sufficiency Is Relation Refinement                                        | FR48                                                                                     |
| Corollary 6.23: Arbitrary Set-Valued Payload Transfer via Failure Tokens                 | FR228, FR227, FR226                                                                      |
| Proposition 6.18: The Admissibility Layer Is Nonempty                                    | FR200                                                                                    |
| Proposition 6.8: Positive Affine Utility Reparameterizations Are Invisible               | FR13, FR16, FR15, FR14                                                                   |
| Proposition 4.1: Binary Pairwise Symmetry Dichotomy                                      | FR120, FR118, FR119                                                                      |
| Proposition 6.20: Bounded-Pattern Predicates Stabilize Above a Finite Action Bound       | FR215, FR216, FR214                                                                      |
| Proposition 6.15: Closure Operations Preserve Exact Certification                        | FR57, FR17, FR18, FR60, FR56, FR59, FR84, FR16, FR55, FR58, FR83, FR15, FR19, FR20, FR85 |
| Proposition 4.2: Decision-Relevant Binary Pairwise Dichotomy                             | FR121, FR123, FR124, FR122                                                               |
| Proposition 3.2: Degenerate Positive Cases                                               | FR1, FR6                                                                                 |
| Proposition 6.1: Exact Certification Depends Only on the Decision Quotient Relation      | FR32, FR33, FR31, FR30, FR37, FR51, FR35, FR34                                           |
| Proposition 6.19: Bounded Distinct Action Profiles Compress to Bounded Actions           | FR206, FR207, FR205, FR209, FR208                                                        |
| Proposition 6.9: Duplicate Actions and Duplicate States Preserve Certification           | FR71, FR68, FR54, FR70, FR56, FR69, FR55, FR73, FR72                                     |
| Proposition 6.7: Explicit Enumeration Is Parameter-Dependent, Not Structural             | FR66, FR67                                                                               |
| Proposition 2.4: Explicit Primitive Basis                                                | FR29                                                                                     |
| Proposition 2.2: Explicit Inventory                                                      | FR26, FR25, FR3, FR28, FR27                                                              |
| Proposition 2.5: All Independent Core Mechanisms Are Nondegenerate                       | FR108, FR106, FR112, FR101, FR105, FR113                                                 |
| Proposition 2.9: Current Positive Interfaces Factor Through Role Classification          | FR4                                                                                      |
| Proposition 6.12: Invariance Alone Is Too Weak                                           | FR22, FR21, FR23                                                                         |
| Proposition 2.6: Weak Tree Ordering Does Not Imply Width One                             | FR115, FR114                                                                             |
| Proposition 3.1: Lifted Cases Introduce No New Primitive                                 | FR5                                                                                      |
| Proposition 6.14: Unrestricted Predicates Trivialize Exact Characterization              | FR145                                                                                    |
| Proposition 4.12: Orbit-Gap Completeness for Exact Classification                        | FR203, FR201, FR202, FR204                                                               |
| Proposition 4.3: Orbit-Gap Template                                                      | FR185                                                                                    |

| <b>Paper claim</b>                                                                    | <b>Lean handle</b>                       |
|---------------------------------------------------------------------------------------|------------------------------------------|
| Proposition 2.8: Parent-Tree Structure Gives Width-One Decompositions                 | FR110, FR100, FR111                      |
| Proposition 6.21: Deterministic Payload Transfer                                      | FR219, FR220, FR222, FR218, FR221, FR217 |
| Proposition 6.11: Quotient Size Is Unbounded Under Realizability                      | FR24, FR38                               |
| Proposition 5.3: The Realized Quotient Is Exactly the Label Range                     | FR39, FR40                               |
| Proposition 6.10: Relabeling Invariance Is Forced by Exact Certification              | FR17, FR18, FR19, FR20                   |
| Proposition 6.26: Exact Semantic Claims Depend Only on Admissible-Output Equivalence  | FR236, FR235                             |
| Proposition 5.2: Every Equivalence Relation Is Optimizer-Realizable                   | FR47, FR46, FR45                         |
| Proposition 6.13: Independent Summary Frameworks Converge on the Same Structural Core | FR178, FR181, FR182, FR180, FR179        |
| Proposition 2.7: Cyclic Dependencies Recover Hardness                                 | FR110, FR100, FR111, FR115               |
| Proposition 6.5: Zero-Distortion Summaries Refine the Optimizer Quotient              | FR177, FR176                             |
| Theorem 4.8: Admissible Collapse Across the Four Obstruction Families                 | FR190                                    |
| Theorem 4.9: Closure-Sound Package No-Go                                              | FR193                                    |
| Definition 6.17: Admissible Normalization Predicate                                   | FR197, FR199, FR198                      |
| Theorem 2.1: Explicit Partition of the Current Positive Landscape                     | FR26, FR25, FR1, FR2, FR28, FR27         |
| Theorem 4.4: Dominant-Pair Admissible No-Go                                           | FR186                                    |
| Theorem 4.6: Ghost-Action Admissible No-Go                                            | FR188                                    |
| Theorem 5.1: Every Labeling Kernel Is Optimizer-Realizable                            | FR7, FR8                                 |
| Theorem 4.5: Margin-Masking Admissible No-Go                                          | FR187                                    |
| Theorem 4.7: Offset Admissible No-Go                                                  | FR189                                    |

*Auto summary: mapped 54/54 (full=54, derived=0, unmapped=0).*